

Using Grocery Data for Credit Decisions

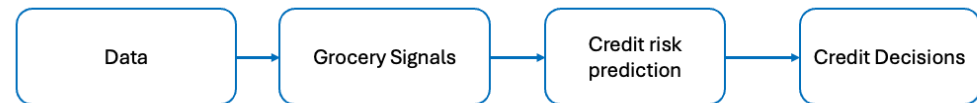
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Summary



30,089 consumers	• WHAT to buy Product choices and spending shares • HOW to buy Regularity, variety	XGBoost + Nested CV • Level 1 Never delinquent vs. ever delinquent • Level 2 Sloppy payer vs. defaulter	Predicted probabilities ↓ Expected payoff ↓ Approval decision
Core Finding:		+7.66 pp AUC unscored consumers	+1.46% payoff unscored consumers

Grocery data create the most value when conventional credit information is sparse.

Research Question

Can grocery-shopping habits provide incremental signals of credit risk, particularly for consumers without traditional credit scores?

- Yes. Features derived from what and how consumers buy provide incremental predictive value, especially when credit scores are unavailable.
- Prediction gains can improve credit approval payoffs when the lender uses grocery data as an additional screening device.

Motivation

- Practical Motivation
 - Many consumers lack traditional credit scores because they have limited or no credit histories, which prevents lenders from accurately evaluating their creditworthiness.
 - Financial institutions are interested in alternative data both to acquire new customers and to extend credit access to underserved consumers.
 - According to the industry managers interviewed by the authors, the main obstacle is not access to granular consumer data, but knowing how to summarize such data into meaningful variables.

Motivation

- Why Grocery Data
 - Grocery products are nondurable necessities, so a large share of consumers make frequent and repeated choices in this domain.
 - Grocery data capture actual consumer choices, rather than stated intentions or self-reported behavior.
 - Consumers who consistently demonstrate "good" behaviors in grocery shopping may also exhibit good financial behaviors, such as paying credit card bills on time.

Contributions

- **Alternative data for credit scoring**

- Prior studies examine mobile phone usage, text, soft information, social networks, digital footprints, verified consumer data, and education and employment histories.
- This paper assesses the information content of **grocery transaction data**, a largely unexplored source of alternative data.

- **Economics of data**

- Prior research studies how data and data sharing generate value through improved prediction and customization.
- This paper studies first-party data sharing across seemingly remote domains: grocery retail and consumer finance.

- **From prediction to decision**

- The paper separates the firm's use of data into **prediction** and **decision** stages.

Data Sources

Data source	Main information	Period
Credit card issuer	Sociodemographics: monthly income, employment status, occupation, and number of dependents	Application records
Credit bureau	Traditional credit scores purchased by the issuer	Available for 50.3% of the final sample
Credit card issuer	Transaction-level spending: date, amount, inflow/outflow, merchant category, and anonymized merchant name	Jan. 2017–Dec. 2018
Credit card issuer	Monthly payment history: delinquency duration and payment status	Jun. 2017–Jun. 2019
Supermarket	Loyalty-card transactions: timestamp, item and brand codes, product hierarchy, quantity, listed and paid prices, transaction, customer, and store identifiers	Jan. 2017–Dec. 2018
Supermarket	Store-level promotions: promotion dates and duration, regular and promotional prices, margin, inventory, and quantity sold	Jan. 2017–Dec. 2018

Sample

- Final sample contains 30,089 consumers
 - at least five supermarket trips in Period 1, no default in Period 1, and removing unresolved final-month delinquencies
- outcomes
 - never-delinquent if they miss no credit card payment (24,315; 81%)
 - sloppy payers if they have at least one one-month delinquency but no two-month delinquency (3,576; 12%)
 - defaulters if they have at least one two-month delinquency (2,198; 7%)

Summary Statistics

Table 1. Consumer Characteristics by Segment

	Never-delinquents (N)	Sloppy payers (S)	Defaulters (D)	Statistical differences ^a
Monthly income, US\$	12,964 (19,851)	13,217 (22,092)	10,704 (19,106)	$N = S > D$
Credit score	719 (59.5)	696 (72.2)	675 (85.4)	$N > S > D$
Monthly credit card spend, US\$	2,199 (3,374)	1,494 (2,523)	1,596 (3,159)	$N > S = D$
Monthly supermarket spend, US\$	491 (426)	459 (410)	376 (370)	$N > S > D$
Number of one-month delinquencies	– –	2.58 (2.82)	1.48 (2.36)	$S > D$
Number of consumers	24,315	3,576	2,198	

Signal 1: What Consumers Buy

Grocery variables

162 food product groups, measured in Period 1

$$G_{ik} = \begin{cases} \text{Whether purchased}_{ik}, \\ \text{Expenditure share}_{ik}. \end{cases}$$

- Consumers who always pay on time tend to purchase healthier and less convenience-oriented food items. In contrast, defaulters are more associated with purchases such as cigarettes, energy drinks, canned meat, and prepared foods.

Signal 2: How Consumers Buy

Shopping Timing

Measures:

- Day-of-week trip entropy
- Time-of-day trip entropy

Interpretation: Higher entropy means shopping trips are less regular across days or time blocks.

Spending Amounts

Measures:

- Variance of log monthly expenditure
- Variance of log per-trip expenditure

Interpretation: Higher variance means grocery spending is less stable.

Items Purchased

Measures:

- Number of unique groups, categories, UPCs, and brands

Interpretation: Higher values indicate broader variety and lower persistence in choices.

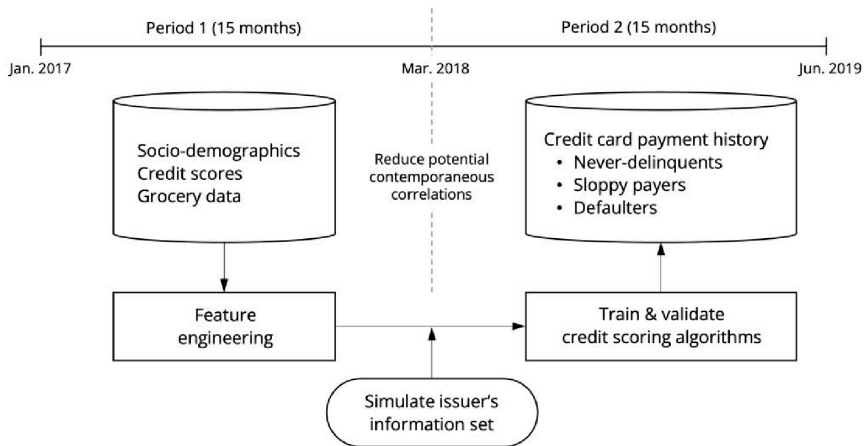
Use of Promotions

Measures:

- Basket/expenditure share of sale UPCs
- $\Pr(\text{buy sale UPC} \mid \text{buy category})$

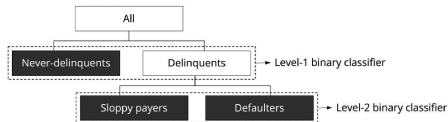
Interpretation: Higher values indicate stronger use of price promotions.

Incremental Predictive Power of Grocery Data



Incremental Predictive Power of Grocery Data

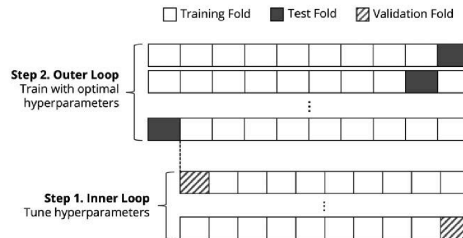
1. Hierarchical Classification



Level 1: Never- vs. ever-delinquent

Level 2: Sloppy payer vs. defaulter

2. Nested Cross-Validation



Inner loop: Hyperparameter tuning

Outer loop: Out-of-sample evaluation

Incremental Predictive Power of Grocery Data

- The predictive gain from grocery data is larger among consumers without credit scores.

Panel A. Unscored consumers			
Out-of-sample AUC			
Baseline information set	Without grocery	With grocery	Δ AUC (pp.)
Level 1: Never vs. ever-delinquents			
Sociodemographics	0.570 (0.027)	0.647 (0.017)	+7.66 pp. (2.59)
Level 2: Sloppy payers vs. defaulters			
Sociodemographics	0.560 (0.031)	0.591 (0.048)	+3.11 pp. (4.50)
Panel B. Scored consumers			
Out-of-sample AUC			
Baseline information set	Without grocery	With grocery	Δ AUC (pp.)
Level 1: Never vs. ever-delinquents			
Sociodemographics	0.568 (0.020)	0.629 (0.015)	+6.07 pp. (1.34)
Sociodemographics + credit scores	0.654 (0.033)	0.679 (0.025)	+2.51 pp. (1.48)
Level 2: Sloppy payers vs. defaulters			
Sociodemographics	0.623 (0.029)	0.630 (0.038)	+0.665 pp. (2.55)
Sociodemographics + cred			+0.359 pp.

From Prediction to Credit Decisions

For applicant i , the model predicts:

$$p_i = (p_i^N, p_i^S, p_i^D),$$

where N , S , and D denote never-delinquent, sloppy payer, and defaulter, respectively.

Expected payoff from approval

$$EV_i = p_i^N \pi_i^N + p_i^S \pi_i^S + p_i^D \pi_i^D,$$

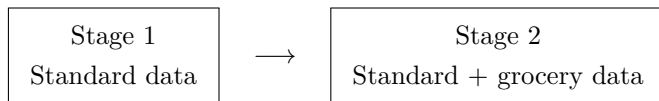
where π_i^j is the issuer's payoff if applicant i is approved and becomes type j .

$$\text{Approve applicant } i \iff EV_i \geq \delta,$$

where δ is the minimum expected-payoff threshold.

From Prediction to Credit Decisions

Two-stage screening mechanism



- The approval threshold δ is chosen to rationalize the observed approval of all sample consumers in Stage 1.
- Grocery data are then used as an additional screening device: applicants with updated $EV_i < \delta$ are rejected.
- Per-person payoff = $\frac{\sum_{i \in \text{Approved}} EV_i}{\# \text{Approved applicants}}$

From Prediction to Credit Decisions

- Grocery data improve the composition of approved applicants by screening out defaulters more aggressively, especially when credit scores are unavailable.

Panel A. Unscored consumers				
Baseline information set	Δ Per-person payoff	Δ Pr(Approve)		
		N	S	D
Sociodemographics	1.46% (1.32)	-2.06 pp. (1.52)	-4.44 pp. (3.62)	-7.16 pp. (5.58)
Panel B. Scored consumers				
Baseline information set	Δ Per-person payoff	Δ Pr(Approve)		
		N	S	D
Sociodemographics	1.48% (0.660)	-2.39 pp. (0.841)	-5.26 pp. (3.10)	-9.69 pp. (3.52)
Sociodemographics + credit scores	0.025% (0.079)	-0.017 pp. (0.036)	-0.142 pp. (0.300)	-0.133 pp. (0.422)

Replication

Panel A. Unscored consumers				
Baseline information set	Out-of-sample AUC		Δ AUC (pp.)	Δ AUC (%)
	Without grocery	With grocery		
	Level 1: Never vs. ever-delinquents			
Sociodemographics	0.570 (0.027)	0.647 (0.017)	+7.66 pp. (2.59)	+13.6% (5.01)
Sociodemographics	Level 2: Sloppy payers vs. defaulters		+3.11 pp. (4.50)	+5.69% (8.26)
	0.560 (0.031)	0.591 (0.048)		

Replication

Panel A. Unscored consumers				
Baseline information set	Without grocery	With grocery	Δ AUC (pp.)	Δ AUC (%)
Level 1: Never vs. ever-delinquents				
Sociodemographics	0.553 (0.037)	0.604 (0.053)	+5.07 pp. (4.44)	+9.26% (8.04)
Level 2: Sloppy payers vs. defaulters				
Sociodemographics	0.479 (0.044)	0.505 (0.108)	+2.56 pp. (12.62)	+6.60% (27.99)

Idea: Public Alternative Data

- 个人的信贷违约是难以获取的数据，但类似的可以做企业的违约或者破产预测
- 裁判文书网中企业的司法行政处罚数据
- 公开招聘数据

Why Is This a *Management Science* Paper?

1. Important managerial and societal problem
2. Novel and uniquely matched proprietary data
3. **From prediction to managerial decisions**

The paper goes beyond AUC improvements and shows whether better predictions change credit approval decisions and lender payoffs.

4. **Clear boundary conditions**

Grocery data are most valuable when traditional information is scarce, but their marginal value declines as credit scores and lender-specific transaction data become available.

Food Items Purchased by Credit Card Segment

entropy indicates

$$Entropy_i = - \sum_{k=1}^K p_{ik} \log(p_{ik}), \quad p_{ik} = \frac{\text{trips in category } k}{\text{total trips}}$$

Multinomial Logit Model

$$\Pr(Y_i = j \mid G_{ik}, X_i) = \frac{\exp(\alpha_j + \beta_j G_{ik} + X_i' \gamma_j)}{\sum_{\ell \in \{N, S, D\}} \exp(\alpha_\ell + \beta_\ell G_{ik} + X_i' \gamma_\ell)}.$$

- $Y_i \in \{N, S, D\}$: never-delinquent, sloppy payer, or defaulter.
- X_i : sociodemographics, credit score, and log total grocery expenditure.
- One regression per food item; results reported as average marginal effects.

Grocery Habits by Credit Card Segment

Grocery habit G_i	Never-delinquents	Sloppy payers	Defaulters
Panel A. Consistency in the timing of shopping			
Day-of-week trip entropy	-0.084*** (0.008)	0.043*** (0.006)	0.041*** (0.005)
Time-of-day trip entropy	-0.046*** (0.007)	0.023*** (0.005)	0.023*** (0.004)
Panel B. Consistency in spending amounts			
Variance of log monthly grocery expenditure	-0.009*** (0.001)	0.006*** (0.001)	0.004*** (0.001)
Variance of log per-trip grocery expenditure	-0.005 (0.004)	0.002 (0.003)	0.003 (0.002)